

Exercise 3: Google App Engine SDK and Web App Development

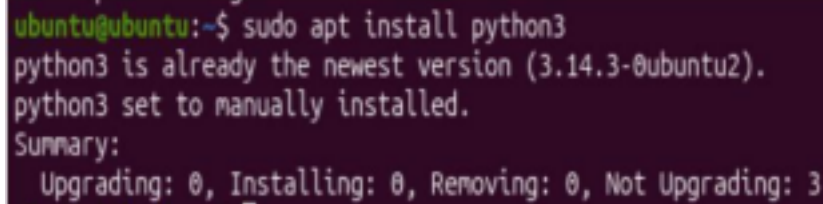
Aim

To install the Google App Engine SDK and develop a Hello World / simple web application using Python.

Procedure

Step 1: Install Google Cloud CLI & Python Ensure

Python 3 is installed on the system.



```
ubuntu@ubuntu:~$ sudo apt install python3
python3 is already the newest version (3.14.3-0ubuntu2).
python3 set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 3
```

Download and install the Google Cloud CLI (contains the modern App Engine SDK) from cloud.google.com/sdk/docs/install:

```
curl -O https://dl.google.com/dl/cloudsdk/channels/rapid/downloads/google-cloud-cli
linuxx86_64.tar.gz
tar -xf google-cloud-cli-linux-x86_64.tar.gz ./google-cloud-sdk/install.sh
```

Step 2: Initialize GCloud & Install Components gcloud

init

Sign in to your Google account when prompted.

Step 4: Create Application Code (main.py) from

```
flask import Flask
app = Flask(__name__)
```

```
@app.route("/") def
hello():
return "Hello World from Google App Engine!"
```

```
if __name__ == "__main__":
app.run(host="127.0.0.1", port=8080, debug=True)
```

Step 5: Create Configuration File (app.yaml)

```
runtime: python39
handlers:
- url: /*
script: auto
```

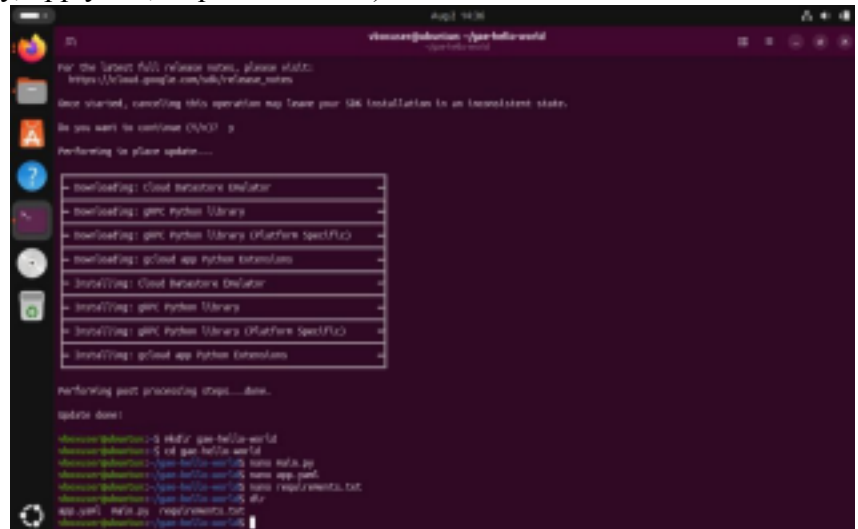
Step 6: Create requirements.txt

```
Flask==2.0.1
```

Output

Files created in project directory (main.py, app.yaml, requirements.txt):

231501152 Shree Varshini A



Result

Thus, the Google App Engine SDK was installed and a simple Hello World web application was developed using Python and Flask.